

MYELOID NEOPLASIA

Type-I interferon-activated NK cells control polycythemia vera in vivo

Johanna Lossa,^{1,2} Tina M. Schnöder,³ Zora Ronge,^{1,2} Nelly Haasch,^{1,2} Katrin Hodapp,^{1,2} Federico Marini,^{2,4} Najla Abassi,^{2,4} Matthias M. Gaida,^{2,5,6} Shin-Jye Lee,⁷ Alina Henrich,⁶ Michaela N. Höhne-Wiechmann,⁶ Carlos Thull Mogollón,⁸ Kristian Schütze,^{1,2} Matthias Klein,^{1,2} Tobias Bopp,^{1,2} Hansjörg Schild,^{1,2} Daniel Sasca,^{2,8,9} Carl Crodel,^{9,10} Florian H. Heidel,^{3,9,11} Steffen Koschmieder,^{9,12,13} Hans Christian Probst,^{1,2,*} Markus P. Radsak,^{2,8,9,14,*} and Sabine Muth^{1,2,*}

¹Institute of Immunology and ²Research Center for Immunotherapy, University Medical Center of the Johannes Gutenberg University, Mainz, Germany; ³Hematology, Hemostasis, Oncology and Stem Cell Transplantation, Hannover Medical School, Hannover, Germany; ⁴Institute of Medical Biostatistics, Epidemiology and Informatics, ⁵Institute of Pathology, and ⁶Translational Oncology, University Medical Center of the Johannes Gutenberg University Mainz gGmbH, Mainz, Germany; ⁷Research and Development Department, PharmaEssentia Corporation, Taipei, Taiwan; ⁸Department of Hematology and Oncology, University Medical Center of the Johannes Gutenberg University, Mainz, Germany; ⁹German Study Group for Myeloproliferative Neoplasms, Germany; ¹⁰Innere Medizin II, Hematology and Oncology, University Hospital Jena, Jena, Germany; ¹¹Leibniz Institute on Aging, Fritz Lipmann Institute, Jena, Germany; ¹²Department of Hematology, Oncology, Hemostaseology and Stem Cell Transplantation, Faculty of Medicine, RWTH Aachen University, Aachen, Germany; ¹³Center for Integrated Oncology Aachen Bonn Cologne Düsseldorf, Aachen, Germany; and ¹⁴Department of Medicine VII, Hematology and Oncology, Donau-Isar-Klinikum, Deggendorf, Germany

KEY POINTS

- NK cells are indispensable for the therapeutic effects of IFN- α in PV.
- Tonic type-I interferon sensing by NK cells attenuates PV disease progression.

Polycythemia vera (PV) is a clonal hematopoietic stem cell disorder resulting in overproduction of erythrocytes. Although interferon alfa (IFN- α) has shown therapeutic efficacy in PV and other myeloproliferative neoplasms (MPN), its precise mechanism of action remains poorly understood. In this study, we identify natural killer (NK) cells as primary immune effectors responsive to IFN- α treatment in PV essential for disease control in vivo. Using a transgenic mouse model of PV, we demonstrate that IFN- α induces the expansion of CD27⁺ NK cells in the bone marrow. In patients with PV or essential thrombocythemia undergoing IFN- α therapy, the frequency of CD56^{bright} NK cells is increased and correlates with the molecular response. Depletion of NK cells abrogated the therapeutic effects of IFN- α . In vitro experiments demonstrate that NK

cells preferentially killed Jak2VF mutant hematopoietic stem and progenitor cells in a tumor necrosis factor- α -dependent manner and independent of IFN- γ or NKG2D. Notably, PV mice depleted of NK cells or lacking type-I interferon receptor on NK cells showed accelerated disease progression in the absence of exogenous IFN- α . This suggests that direct sensing of basal levels of type-I interferons by NK cells is essential for attenuating disease progression, emphasizing a critical role for NK cells in immune surveillance of MPN. These findings offer new insights into type-I interferon-mediated immune modulation in MPN and highlight the potential of NK cell activation to improve therapeutic outcomes.

Introduction

Myeloproliferative neoplasms (MPN) are clonal hematopoietic stem cell (HSC) disorders resulting in an overproduction of mature myeloid cells in the peripheral blood and bone marrow (BM). The Jak2V617F mutation is highly prevalent in polycythemia vera (PV) and leads to constitutively active Janus kinase 2 (JAK2) signaling,¹ which results in excessive myeloid cell proliferation including overproduction of erythrocytes.²⁻⁵ The disease is accompanied by thrombophilia and chronic inflammation that promotes clonal evolution and disease progression,⁶ ultimately leading to tumor immune escape.⁷

The discovery of the Jak2V617F (Jak2VF) point mutation as the major driver of PV in 2005 by several groups²⁻⁵ has triggered the development of JAK inhibitors, which may relieve PV-induced symptoms and splenomegaly. However, standard treatments including JAK inhibitors, phlebotomy, or chemotherapy do not eradicate malignant HSCs and they are thus not curative.⁸ In contrast, interferon alfa (IFN- α) has been shown to be an effective treatment in all MPN, inducing long-term remissions^{9,10} and preventing disease progression.¹¹ Although a direct effect of IFN- α was seen on Jak2VF-positive HSCs,^{12,13} the mechanisms of action of interferon in MPN are only poorly understood, particularly to which extent

nonclonal cells such as immune cells contribute to the efficacy of IFN- α . In general, type-I interferons are central regulators of innate as well as adaptive immune responses and are produced in large amounts upon viral infections. Many cell types including immune as well as stromal cells express the receptor for type-I interferon and can therefore directly respond to type-I interferon activating the JAK/STAT signaling pathways¹⁴ and ultimately resulting in the expression of interferon-stimulated genes.¹⁵ Thus, treatment with type-I interferon can have pleiotropic effects.¹⁶

In this work, we use a transgenic knockin mouse model of PV¹⁷ to study the effects of type-I interferon treatment on the immune system. We describe a marked expansion and activation of CD27⁺ natural killer (NK) cells in the BM of mice upon IFN- α treatment and an increase of CD56^{bright} NK cells in patients with PV or essential thrombocythemia (ET) receiving IFN- α therapy. Importantly, the therapeutic effects of IFN- α were abrogated by NK cell depletion, demonstrating that NK cells are critical effector cells in IFN- α therapy of PV. We further show that NK cells preferentially kill Jak2VF-mutant HSC in a tumor necrosis factor- α (TNF- α)-dependent manner.

Taken together, we show that IFN- α -mediated NK cell activation is a key mechanism to control excessive cell proliferation in PV, targeting preferentially malignant HSCs. Our results deepen the understanding of the well-known therapeutic efficacy of IFN- α in MPN.

Materials and methods

Detailed lists of reagents and their suppliers are given in the supplemental Materials, available on the *Blood* website. Additional methods are described and provided in the supplemental Methods.

Samples

Peripheral blood samples of patients with MPNs and healthy volunteers were analyzed, and basic patient characteristics were recorded as summarized in supplemental Table 1. Detailed information about characteristics of patients with PV/ET are displayed in supplemental Tables 2-5. All work was carried out in accordance with the Declaration of Helsinki and approved by the Landesärztekammer Rheinland-Palatine ethics committee according to the institutional guidelines and were conducted with the understanding and informed consent from all participants.

C57BL/6J mice were purchased (Charles River). Jak2V617F/+ (B6.N.129S6(SJL)-Jak2^{tm1.2Ble/Amly})¹⁷; VavCre (Tg(Vav1-cre)1Cgp)¹⁸ double-transgenic mice (Jak2VF), CD45.1 (B6.SJL-Ptprc^a Pepc^b/BoyJ)¹⁹ mice, Ncr1-Cre (B6.Cg-Ncr1^{tm1.1(cre)Viv/Orl})²⁰ mice, Ifnar1fl/fl (B6(Cg)-Ifnar1^{tm1.2EEs/J})²¹ and Ifnar1^{-/-} (B6.129S2-Ifnar1^{tm1Agt} (N+20 B6))²² mice have been described previously. All mice were bred and maintained under specific-pathogen-free conditions in the central animal facility of the Johannes Gutenberg-University, Mainz, Germany. Generation of BM chimeric mice has been described previously.²³ Experiments were performed with age- and sex-matched mice and were conducted with permission of the Landesuntersuchungsamt Rheinland-Palatinat.

Treatment of mice

In all experiments involving IFN- α therapy, mice were treated with 30 ng murine ropeginterferon- α -2b (ropeg-IFN- α ; mP1101, PharmaEssentia) administered subcutaneously twice per week for 4 weeks, or with phosphate-buffered saline as control. mP1101 is a monopegylated IFN- α with an extended half-life that has been shown to exert both immunomodulatory effects and direct activity on murine cancer cells.²⁴ It was used in all experiments to avoid the burden of daily injections of IFN- α to experimental animals and to more closely reflect the clinical use of pegylated-IFN- α . NK cells and CD4⁺ and CD8⁺ T cells were depleted by IV injection of 500 μ g antibodies against mouse NK1.1 (clone PK136), CD4 (clone GK1.5), or CD8 (YTS169.4), once a week.

Antibodies and flow cytometric analysis

Antibodies against mouse NK1.1 (clone PK136), CD4 (GK1.5), and CD8 (YTS169.4) were purified from hybridoma culture supernatant using protein G Sepharose (GenScript). Single-cell suspensions were stained as previously described²³ and analyzed on BD FACSymphony (Becton Dickinson). Data were analyzed using FlowJo Software (version 10.8.1, Tree Star).

Generation of Jak2WT-HOXA9 and Jak2VF-HOXA9 LSK cells

LSK (Lin⁻Sca⁺Kit⁺) cells were sorted (FACSARIA, Becton Dickinson) from the BM of either Jak2WT or conditional Jak2VF mice.¹⁷ LSK cells were retrovirally transduced with MSCV-IRES-GFP_HOXA9, as previously described,²⁵ to promote unlimited proliferation of LSK cells. Green fluorescent protein-positive cells were sorted and afterward cultured in RPMI-1640 medium supplemented with 10% fetal bovine serum, 2mM L-glutamine, 50 U/mL penicillin, 50 mg/mL streptomycin, 6 ng/mL interleukin-3 (IL-3), and 10 ng/mL IL-6.

Ex vivo restimulation of NK cells

Isolated BM cells or peripheral blood mononuclear cells (PBMCs) from Jak2VF or Jak2-wild-type (WT) chimeric mice or PBMCs from patients with PV were restimulated for 7 hours at 37°C with 0.05 μ g/mL PMA (phorbol 12-myristate 13-acetate) and 0.5 μ g/mL ionomycin (both Sigma) in Iscove modified Dulbecco medium supplemented with 10% fetal bovine serum, 2mM L-glutamine, 50 U/mL penicillin, 50 mg/mL streptomycin, and 40 ng/mL murine or human IL-15 (both PeproTech). Brefeldin A (5 μ g/mL) was added during stimulation to inhibit cytokine secretion. Cells were subsequently fixed with 2% paraformaldehyde and stained intracellularly for TNF- α .

Statistical analysis

Statistical analysis was performed in GraphPad Prism (version 10.4.1) and is detailed in the figure legends.

Results

IFN- α therapy promotes expansion of cytokine-producing NK cell subsets in a murine model and human patients with MPN

To investigate the contribution of pleiotropic immune activation by type-I interferon to the efficacy of IFN- α therapy in PV, we made use of a well-established mouse model for PV (Jak2VF mice).¹⁷ To delay disease onset and mirror human

disease, we generated mixed BM chimeric mice by transplanting a mixture of BM from Jak2VF mice and BM from WT mice (CD45.1 congenic) into lethally irradiated C57BL/6J mice (CD45.2; [Figure 1A](#)). In these chimeras, only a small proportion of HSCs carry the Jak2VF mutation reflecting the natural course of the disease in human patients. As a control, mice underwent transplantation with a mixture of BM from WT mice (single transgenic Jak2VF mice) and CD45.1 congenic WT mice. Four weeks after reconstitution, when onset of PV in Jak2VF chimeras had become apparent by elevated blood cell counts ([Figure 1B](#)), we started treatment with ropeg-IFN- α . In accordance with previous work,^{12,26} IFN- α treatment reduced disease signs in Jak2VF mixed chimeras as indicated by reduced hematocrit, white blood cell count, lymphocyte count, and hemoglobin levels compared with Jak2VF chimeras treated with vehicle ([Figure 1C](#)), although 4 weeks of ropeg-IFN- α treatment did not reduce the allelic burden, that is, frequency of Jak2V617F⁺ cells in the peripheral blood (supplemental [Figure 1A](#)). To investigate the effects of systemic IFN- α treatment on immune cell populations in the BM, we analyzed frequencies of various immune cells of Jak2VF and control chimeric mice with or without IFN- α therapy by high dimensional flow cytometry. Identification of immune cell populations using flow self-organizing maps and visualization of these populations using the uniform manifold approximation and projection²⁷ dimensionality reduction algorithm revealed no major differences between Jak2VF chimeras and Jak2WT controls ([Figure 1D](#)). Compared with control chimeras, Jak2VF chimeras showed increased frequencies of Ly6C^{high} monocytes and plasmacytoid dendritic cells (DCs) and reduced frequencies of eosinophils (supplemental [Figure 1B-H](#)). IFN- α treatment resulted in increased frequencies of CD4⁺ T cells and plasmacytoid DCs (supplemental [Figure 1F,H](#)) and markedly increased frequencies and absolute numbers of NK cells ([Figure 1E](#); supplemental [Figure 1K](#)) in both Jak2VF and control chimeras, indicating an effect of IFN- α on NK cells independent of the presence of mutant cells. We observed a tendency toward increased NK cell frequencies in the blood, however this only reached significance in the WT chimeras (supplemental [Figure 1K](#)). Notably, most NK cells originate from the Jak2WT (CD45.1) clone, whereas only a minority derive from the Jak2F-positive clone (CD45.2), reflecting the mixture of BM cells used for BM reconstitution. The ratio of NK cells originating from WT and malignant clones remained unchanged under IFN- α therapy (supplemental [Figure 1L](#)). NK cells can be subdivided into 4 subsets based on their expression of CD11b and CD27, which constitute different NK maturation steps.²⁸ We detected no differences in the frequencies of NK cell subsets among Jak2VF and Jak2WT mixed chimeric mice, but IFN- α therapy significantly increased the proportion of immature, CD27⁺ NK cells in the BM ([Figure 1F](#)). Histological analysis of BM sections of a few representative Jak2VF chimeric mice revealed the characteristic increase in loose clusters of megakaryocytes associated with PV,²⁹ which were attenuated after treatment with IFN- α , demonstrating the therapeutic effect of the intervention ([Figure 1G](#)). Immunohistochemical staining for NK1.1 revealed heterogenous distribution of NK cells in the BM without obvious cluster formation ([Figure 1H](#), arrows), with NK cell numbers increasing upon IFN- α therapy.

To transfer our findings to the human context, we analyzed PBMC from patients with PV. To increase statistical power and due to the close clinical and molecular similarities of ET and PV, in our analysis, we also included samples from patients with ET alongside samples from patients with PV. All included patients with PV/ ET carried the Jak2V617F mutation. Patients were grouped according to treatment regimens: IFN- α , Jak inhibitor, hydroxyurea, and phlebotomy or no treatment ("watch and wait"). Samples from myelofibrosis (MF), further classified into primary MF, prefibrotic primary MF, and post-PV/ET MF, were included as a distinct subtype of MPN. Healthy individuals served as baseline controls. IFN- α treatment was administered based on standard clinical indications rather than predefined experimental criteria. Importantly, comparison of baseline characteristics, including cohort size; median age and sex distribution (supplemental [Table 1](#)); as well as clinical characteristics, including leukocyte counts, Jak2VF allele burden, IFN- α dosage, duration of therapy, and previous IFN- α treatments (supplemental [Tables 2-5](#)) did not reveal obvious confounders between treatment groups. Flow cytometric analysis revealed no differences in the frequency or absolute number of total CD56⁺ NK cells across all MPN subgroups ([Figure 2A](#); supplemental [Figures 2 and 3A](#)). Similarly, there were no changes in the CD56^{dim} NK cell subset, which comprises most cytotoxic NK cells in the blood ([Figure 2B](#); supplemental [Figure 3B](#)). However, consistent with a previous study,^{30,31} we observed a selective increase in CD56^{bright} NK cells in patients with PV or ET receiving IFN- α therapy ([Figure 2C-D](#); supplemental [Figure 3C](#)) that was independent of IFN- α dose, age of the patient, and concurrent statin or metformin treatment (supplemental [Figure 3D-F](#)). The CD56^{bright} NK cell subset, which corresponds functionally to murine CD27⁺ NK cells,³² increased in PV mice after IFN- α treatment ([Figure 1F](#)). Importantly, the increase in frequency of CD56^{bright} NK cells positively correlates with the molecular response to IFN- α therapy ([Figure 2E](#)), supporting a clinical significance of these observed changes. In contrast, CD56^{bright} NK cell levels did not correlate with patient age or IFN- α dosage. Also, CD56^{bright} NK cells expressed higher levels of activation markers NKG2D, NKG2A, and NKG2C than CD56^{dim} cells across all patient groups and independent of IFN- α treatment ([Figure 2F](#); supplemental [Figure 3G](#)).

Taken together, these results indicate that IFN- α therapy promotes the expansion of immature NK cell subsets in both PV murine model and human patients with PV receiving IFN- α therapy.

Therapeutic effects of IFN- α in PV mice are mediated by NK cells

The increased frequencies and total numbers of NK cells upon IFN- α therapy prompted us to explore their contribution to the therapeutic effects of IFN- α in PV. Therefore, we depleted NK cells during IFN- α therapy of Jak2VF mixed-BM chimeric mice, using NK1.1-specific antibodies ([Figure 3A](#)). As before, Jak2VF chimeras developed PV as indicated by hematocrit levels of >60% both 4 and 8 weeks after reconstitution ([Figure 3B](#)). Four weeks of IFN- α treatment significantly reduced hematocrit levels, however this effect of IFN- α therapy was lost upon

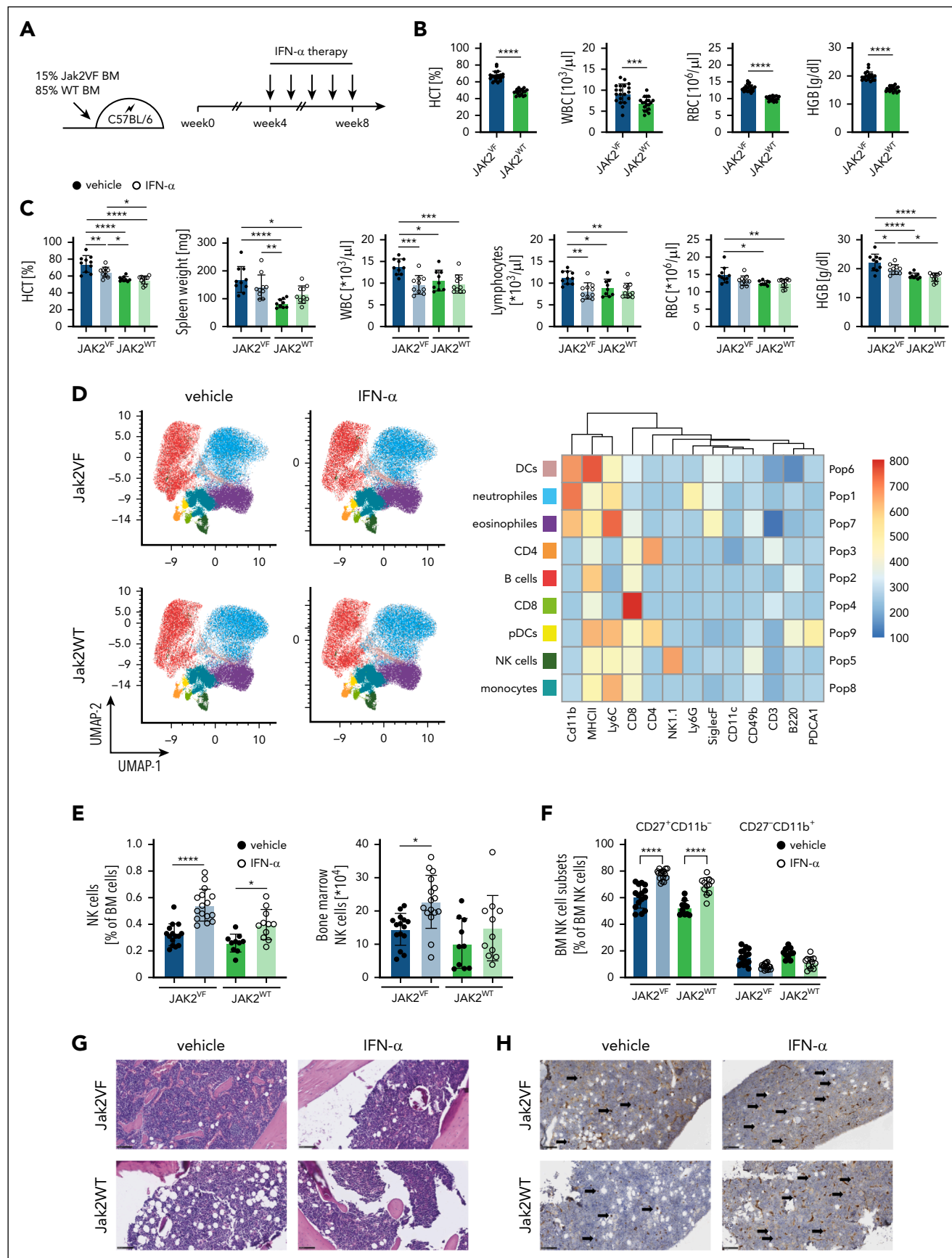


Figure 1.

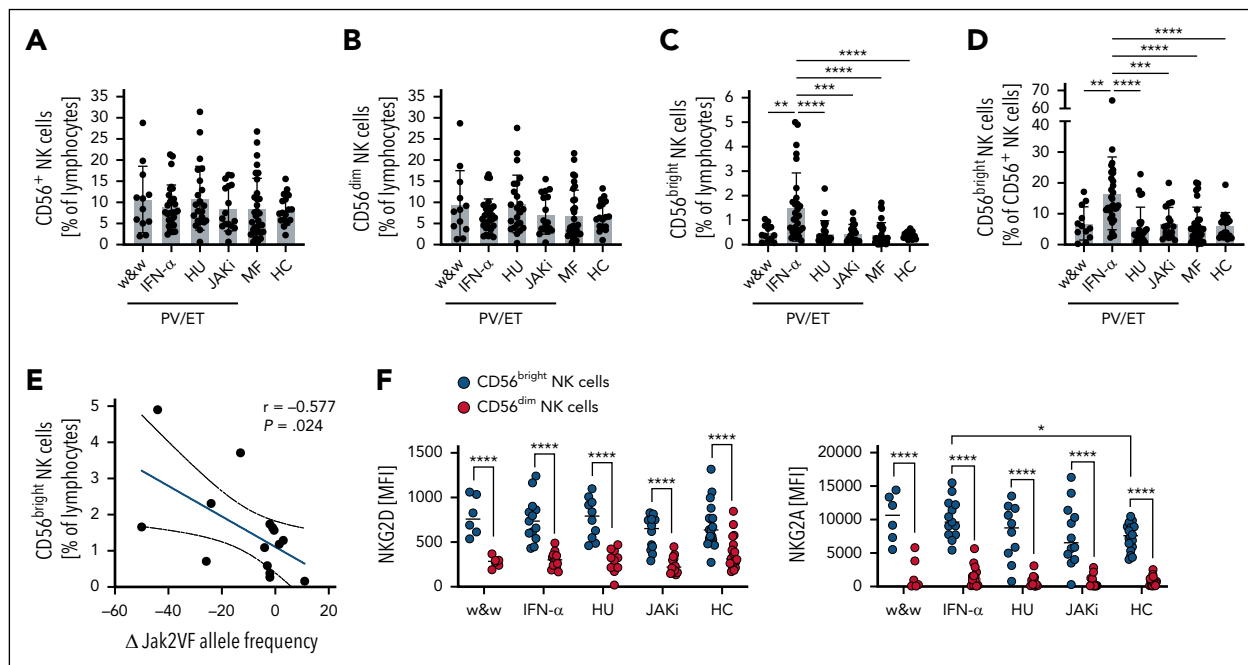


Figure 2. Increased frequencies of peripheral blood CD56^{bright} NK cells in patients with PV or ET receiving IFN-α therapy correlate with the molecular response. PBMCs were isolated from patients with PV and ET undergoing different treatment regimens: no treatment (watch and wait [w&w]; n = 12), HU (n = 23), IFN-α therapy (n = 29), or JAKi therapy (n = 17). Additionally, PBMCs were analyzed from patients with MF receiving HU, JAKi, or IFN-α therapy (n = 32), and from HC (n = 18). Flow cytometry was used to assess the percentage of total CD56⁺ NK cells (A), CD56^{dim} NK cells (B), and CD56^{bright} NK cells (C) among all lymphocytes, and CD56^{bright} NK cells of all CD56⁺ NK cells (D). Data are presented as mean ± SD (horizontal bars). Statistical significance was determined by ordinary 1-way ANOVA followed by the Tukey multiple-comparison test. ****P ≤ .0001; ***P ≤ .001; **P ≤ .01; *P ≤ .05. (E) CD56^{bright} NK cells from patients with PV/ET receiving IFN-α therapy, shown in relation to the change in Jak2V617F allelic burden (difference between Jak2V617 allele frequency (%) at the time of analysis minus baseline frequency measured at treatment initiation or within 12 months before treatment start). Data are presented as scatter dot plots with a simple linear regression (blue line). Black lines indicate the 95% confidence interval. Pearson correlation coefficient was $r = -0.577$ ($P = .024$). (F) Median fluorescence intensity (MFI) of NKG2D and NKG2A on CD56^{bright} and CD56^{dim} NK cell subsets from PBMCs of patients with PV/ET and HC. Data are presented as scatter dot plots with a horizontal line indicating the median. Statistical significance was determined by 2-way ANOVA followed by the Šidák multiple-comparison test. ****P ≤ .0001; ***P ≤ .001; **P ≤ .01; *P ≤ .05. HC, healthy controls; HU, hydroxyurea; JAKi, JAK inhibitor.

concomitant depletion of NK cells (Figure 3B). Furthermore, the decrease in spleen weight, white blood cell counts, and red blood cell counts achieved by IFN-α therapy in PV mixed-chimeric mice was lost upon NK cell depletion during therapy (Figure 3C-E). Depletion of either CD4⁺ T cells or CD8⁺ T cells did not affect the efficacy of IFN-α therapy (supplemental Figure 5A-B). Collectively, these data show that NK cells but not T cells are essential for the therapeutic effect of IFN-α therapy against PV.

IFN-α therapy in PV mice changes gene expression and phenotype of BM NK cells

To investigate how NK cells contribute to therapeutic effects of IFN-α in PV mice, we analyzed NK cell transcriptomes by bulk-RNA sequencing. Gene expression was compared in WT NK cells isolated from the BM of Jak2VF mixed chimeras. In addition

to the expected upregulation of type-I interferon response genes, gene sets associated with cell cycle regulation were enriched, indicating enhanced NK cell proliferation in the BM (supplemental Figure 6A). This is consistent with the increased NK cell frequency and number measured by flow cytometry (supplemental Figure 6B; Figure 1E). However, Gene Ontology terms associated with NK cell activation and function were not enriched (supplemental Figure 6C). The inhibitory receptor genes *Klrc1* and *Klrb1b*, encoding NKG2A and NKR-P1b, were downregulated after IFN-α treatment (supplemental Figure 6D). Although expression of most NK cell effector cytokines and chemokines was unchanged, granzyme expression increased, suggesting modest enhancement of effector function.

Phenotypic analysis of BM NK cells from Jak2VF and Jak2WT chimeric mice on the single-cell level by flow cytometry,

Figure 1. Administration of ropeg-IFN-α in Jak2VF mice reduces PV symptoms and increases frequencies and total numbers of NK cells in the BM. (A) Experimental protocol for IFN-α therapy of PV mice. Mixed BM chimeric mice (BMC) were generated by transplanting lethally irradiated C57BL/6 mice with a mixture of 15% VavCre × Jak2V617F (Jak2VF) and 85% WT (CD45.1) BM (Jak2WT); or, as control, 15% WT (VavCre) and 85% WT (CD45.1) BM (Jak2WT). Starting 4 weeks after reconstitution, mice were treated with murine ropeg-IFN-α-2b twice a week over 4 weeks. (B) Hematocrit (HCT), white blood cell count (WBC), red blood cell count (RBC), and hemoglobin (HGB) 4 weeks after BM transplantation in Jak2VF and Jak2WT mixed chimeric mice. (C) Spleen weight, HCT, WBC, lymphocyte numbers, RBC, and HGB in the blood of Jak2VF BMC or of Jak2WT control chimeras after 4 weeks of IFN-α or control treatment. (D) Identification of immune cells in the BM after 4 weeks of IFN-α or control treatment by high dimensional flow cytometry and flow self-organizing maps. Visualization of immune cell populations using UMAP dimensionality reduction algorithm. (E) Frequencies and total numbers of NK cells in BM of Jak2VF BMC or of Jak2WT controls. (F) Frequencies of CD27⁺CD11b⁺ and CD27⁺CD11b⁺ NK cell subsets of all NK cells in the BM. (G) Hematoxylin and eosin stain and (H) immunohistochemical identification of NK cells using anti-mouse NK1.1 as primary antibody in BM sections of Jak2VF and Jak2WT BMC with or without IFN-α therapy. Horizontal bars represent mean ± standard deviation (SD). Statistical significance was determined by the Student t test (B) or 1-way analysis of variance (ANOVA) followed by the Tukey multiple-comparison test. ****P ≤ .0001; ***P ≤ .001; **P ≤ .01; *P ≤ .05. pDCs, plasmacytoid DCs; UMAP, uniform manifold approximation and projection.

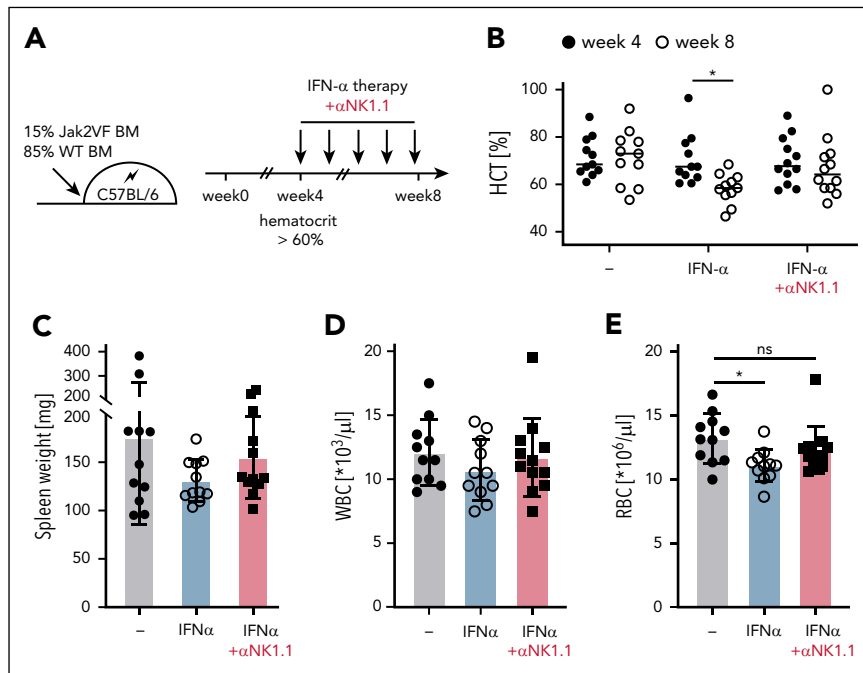


Figure 3. Depletion of NK cells reversed reduced disease symptoms in IFN- α -treated PV BM chimeric mice. (A) Experimental protocol for IFN- α therapy and NK cell depletion of PV mice: BMC were generated by transplanting lethally irradiated C57BL/6 mice with a mixture of 15% VavCre $\times$ Jak2V617F (Jak2VF) and 85% WT (CD45.1) BM (Jak2VF). Starting 4 weeks after reconstitution, when HCT levels exceeded 60%, mice were treated with ropeg-IFN- α -2b twice a week over 4 weeks. In the indicated groups, NK cells were concomitantly depleted by weekly injection with an antibody against NK1.1 (B) HCT before and after treatment with IFN- α or IFN- α + α -NK1.1 (C) Spleen weight of Jak2VF BMC or of Jak2WT control chimeras after IFN- α or IFN- α + α -NK1.1 treatment. (D) WBC in the blood of Jak2VF chimeric mice after IFN- α or IFN- α + α -NK1.1 treatment. (E) RBC in the blood of Jak2VF chimeric mice after IFN- α or IFN- α + α -NK1.1 treatment. Horizontal bars represent mean $\pm$ SD; $n = 11$ to 12. Statistical significance was determined with 2-way ANOVA and the Šidák multiple-comparison test (panel B) or 1-way ANOVA and the Tukey multiple-comparison test (panel E); * $P \leq .05$. ns, not significant.

revealed increased expression of the activating receptor NKG2D after IFN- α therapy, independent of genotype and higher in CD27 $^{+}$ NK cells (Figure 4A). Expression of the inhibitory receptor NKG2A was unchanged but also higher in CD27 $^{+}$ NK cells (Figure 4B), and protein levels of both receptors did not reflect mRNA expression (supplemental Figure 6D), suggesting posttranscriptional regulation. IFN- α also increased Ly49C/F/I/H expression on CD27 $^{+}$ NK cells (Figure 4C). Among NK cells ligands on LSK cells, Rae-1 expression, which binds to NKG2D and enhances NK cell activation, was increased after IFN- α therapy again in PV and control mice (Figure 4D). However, at least after 4 weeks of IFN- α treatment, Qa-1b and major histocompatibility complex I remained unchanged (Figure 4E-F). Together, these findings indicate that IFN- α primarily expands NK cells and induces a moderately more activated phenotype independent of the Jak2VF mutation.

NK cells selectively kill malignant Jak2VF HSCs

Our data indicate that NK cells contribute for the therapeutic effect of IFN- α in PV, but whether NK cells directly kill malignant stem cells remained unclear. To address this, we established an in vitro killing assay using ex vivo NK cells from WT mice. As target cells, we used Lin $^{-}$ Sca1 $^{+}$ Kit $^{+}$ hematopoietic stem and progenitor cells (HSPCs) isolated from either PV or WT mice and retrovirally transduced with HOXA9 to generate immortalized cell lines. NK cells were cocultured at different effector-to-target ratios with these Jak2VF or WT HOXA9 transduced HSPCs. NK cells efficiently killed HSPCs in an effector-to-target-dependent manner, reaching ~40% killing

of Jak2VF cells at the highest ratio (12.5:1). Strikingly, Jak2VF HSPCs were significantly more susceptible to NK cell-mediated killing than WT cells (Figure 5A). Despite upregulation of NKG2D and Rae-1, functional blocking of NKG2D or IFN- γ did not affect selective killing (Figure 5B-C). In contrast, TNF- α blockade markedly reduced killing and abolished preferential targeting of Jak2VF HSPCs (Figure 5D). Consistently, soluble TNF- α preferentially reduced viability of Jak2VF HSPCs compared with WT HSPCs (Figure 5E). Separation of NK cells and target cells in a trans-well system abrogated killing, indicating a cell contact-dependent mechanism (Figure 5F). Together, these results indicate that NK cells selectively kill Jak2VF stem cell through a TNF- α - and cell contact-dependent mechanism. We found equal levels of expression of both TNF receptor 1 and TNF receptor 2 on BM LSK cells from Jak2VF and Jak2WT mice (Figure 5G), thus either receptor might be involved in NK cell-mediated killing. Interestingly, expression of both receptors was upregulated after treatment of mice with ropeg-IFN- α . Staining mouse and human NK cells for TNF- α , which quantifies both intracellular and membrane-bound TNF- α , revealed a stronger TNF- α expression on the CD27 $^{+}$ NK subset in the BM and blood in mice and a tendency toward a higher expression in CD56 $^{\text{bright}}$ NK cells in the blood of patients with PV. TNF- α expression on BM NK cells in Jak2VF mice was enhanced by IFN- α treatment (Figure 5H-I). Taken together, these data suggest that in addition to increasing the number of NK cells, IFN- α treatment might also increase the sensitivity to TNF- α -mediated killing by inducing TNF receptors on HSCs and enhancing TNF- α expression on immature NK cells.

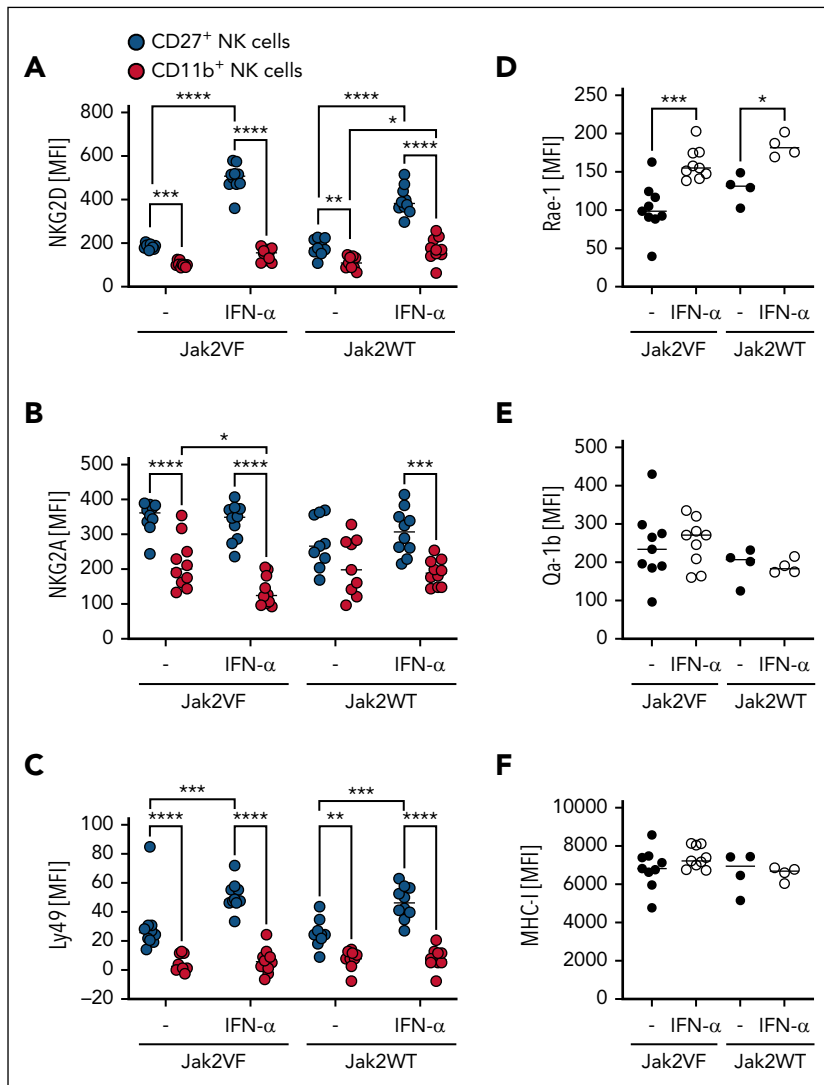


Figure 4. Phenotypic changes of surface receptor expression on NK cell subsets and HSCs from Jak2VF and Jak2WT BM chimeric mice upon IFN- α therapy. (A-C) MFI of NGK2D (A), NGK2A (B), and Ly49 C/F/I/H (C) on CD27⁺ and CD11b⁺ NK cell subsets from the BM of Jak2VF or Jak2WT BM chimeric mice after 4 weeks of IFN- α therapy (n = 9-10). (D-E) MFI of Rae-1 (D), Qa-1b (E), and MHC-I (F) of LSK cells from BM of Jak2VF or Jak2WT BM chimeric mice after 4 weeks of IFN- α treatment or without (n = 4-9). Data are presented as scatter dot plots with a horizontal line indicating the median. Statistical significance was determined with 2-way ANOVA followed by the Šidák multiple-comparison test (panels A-C), or ordinary 1-way ANOVA followed by the Tukey multiple-comparison test (panels D-F). ****P ≤ .0001; ***P ≤ .001; **P ≤ .01; *P ≤ .05. MHC-I, major histocompatibility complex I.

NK cells directly sense steady-state levels of type-I interferons in PV mice and slow disease progression

Type-I interferon are immunostimulatory signals that are rapidly produced upon viral infection and induce multifaceted immune responses.³³ Because the receptor for type-I interferon is expressed on nearly all cell types, type-I interferon can be sensed directly by various immune cells. Because our data indicate that NK cells directly kill Jak2VF-mutated HPSC clones and the therapeutic effect of IFN- α in PV is dependent on NK cells, we wondered whether direct sensing of type-I interferon by NK cells is essential in this context. Therefore, we used Ncr1-Cre mice in combination with *Ifnar^{fl/fl}* mice that lack INF- α receptor (*Ifnar1*) expression on all NK cells. We generated mixed BM chimeric mice of 85% BM of Ncr1CreX*Ifnar^{fl/fl}* and 15% *Ifnar1^{-/-}* Jak2VF, that are deficient for IFNAR1 expression on all cells. In the resulting chimeric mice, 85% of immune cells are WT and therefore express IFNAR1, whereas all NK cells are IFNAR1 deficient and cannot sense type-I interferon directly. As control, mixed chimeric mice of 85% WT (*Ifnar^{fl/fl}*) and either 15% BM of *Ifnar1*-deficient Jak2VF mice or 15% BM of

Ifnar1-proficient Jak2VF mice were generated. As seen before (Figure 1C; Figure 3B), 4 weeks upon reconstitution, both control chimeric mice developed PV-like disease indicated by elevated hematocrit values of ~60% (Figure 6A). However, mice in which solely NK cells cannot sense type-I interferon showed advanced disease progression, emphasized by significantly enhanced hematocrit in comparison with both control PV chimeric mice. These data suggest that already steady-state levels of type-I interferon, which are produced under noninflammatory conditions,³⁴ induce NK cell activity, that targets the malignant stem cells and restrains development of PV. Supporting this type-I interferon-dependent NK cell-mediated control of PV development, we observed a dramatic exacerbation of disease upon NK cell depletion in Jak2VF chimeric mice (Figure 6B).

Discussion

The use of IFN- α 2 in the treatment of MPN has been explored extensively in several clinical studies.³⁵ These studies demonstrated therapeutic efficacy, including normalization of

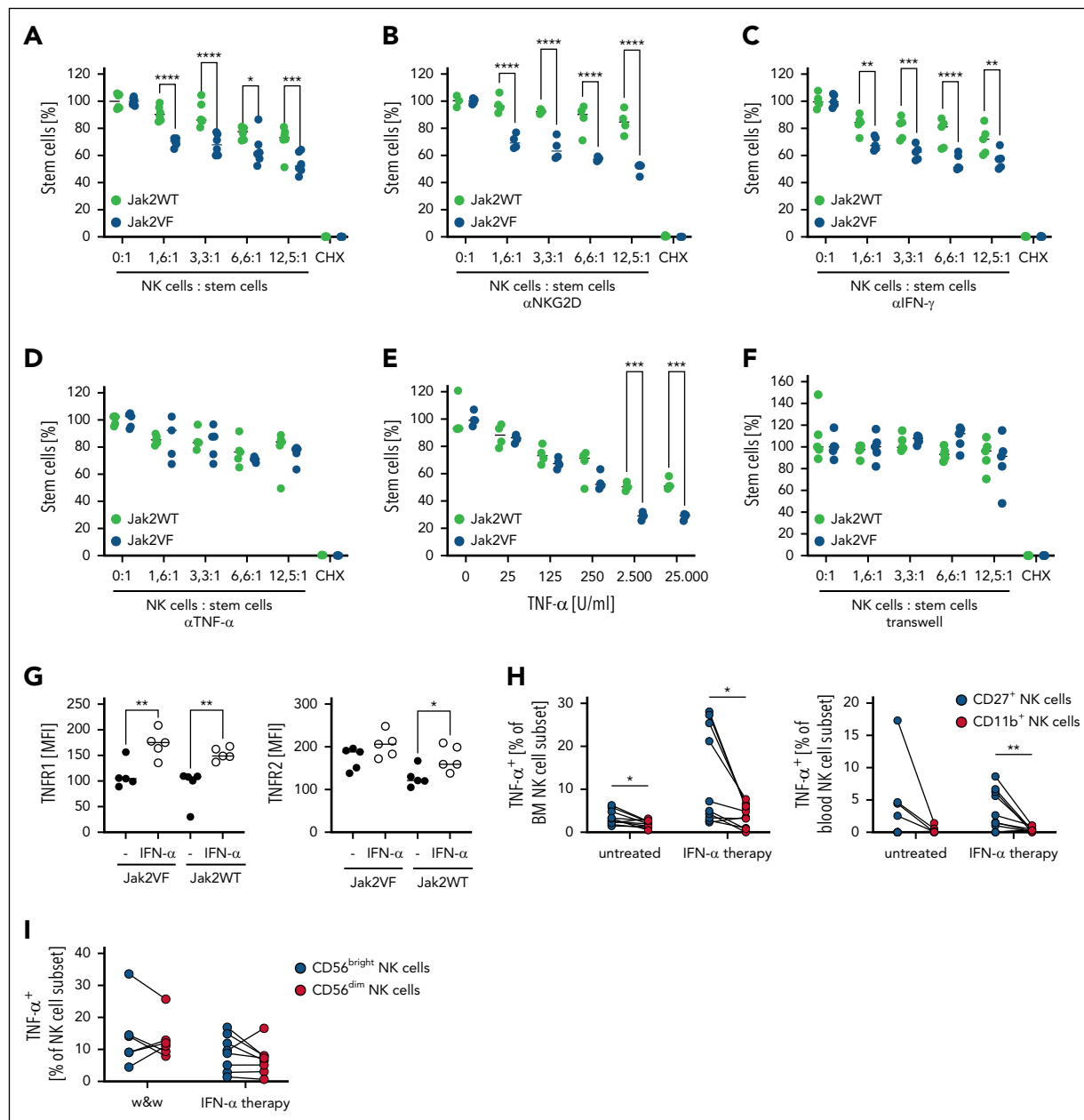


Figure 5. NK cells directly kill malignant stem cells in a TNF- α -dependent manner. (A) Jak2^{VF}HOXA9 LSK (Jak2VF, blue circles) cells or Jak2^{WT}HOXA9 LSK (Jak2WT, green circles) cells were cocultivated for 15 hours with the indicated ratios of WT NK cells. Number of viable Jak2VF or Jak2WT cells was determined by flow cytometry, and their frequencies were calculated by setting the survival rate of the respective stem cells cultured without NK cells to 100%. Treatment with CHX was used as positive control; (B) 10 μ g/mL anti-mouse anti-NKG2D antibody was added to each coculture; (C) 10 μ g/mL anti-mouse anti-IFN- γ antibody was added to each coculture; (D) 10 μ g/mL anti-mouse anti-TNF- α antibody was added to each coculture. (E) Jak2^{VF}HOXA9 (Jak2VF, blue circles) cells or Jak2^{WT}HOXA9 LSK (Jak2WT, green circles) cells were cocultivated for 15 hours in the presence of the indicated amounts of TNF- α and HOXA9 of Jak2VF or Jak2WT cells were determined as in panel A. (F) NK cells and Jak2^{VF}HOXA9 (Jak2VF, blue circles) cells or Jak2^{WT}HOXA9 LSK (Jak2WT, green circles) cells were separated by a permeable transmembrane. (G) MFI of TNFR1 and TNFR2 on LSK cells. BM cells have been isolated from lethally irradiated mice that had been reconstituted with 100% Jak2VF or 100% Jak2WT BM and treated with IFN- α for 48 hours or with phosphate-buffered saline. (H) TNF- α expression of CD27⁺ and CD11b⁺ NK cells from the BM or blood of Jak2VF chimeric mice with or without IFN- α therapy was determined by flow cytometry after restimulation with PMA (phorbol 12-myristate 13-acetate)/ionomycin for 7 hours. (I) TNF- α expression of CD56^{bright} and CD56^{dim} NK cells from PBMCs of patients with PV/ET under w&w (n = 6) or IFN- α therapy (n = 8) was determined by flow cytometry after restimulation with PMA/ionomycin for 7 hours. Statistical significance was determined with 2-way ANOVA followed by the Šidák multiple-comparison test (panels A-F), or 1-way ANOVA followed by the Tukey multiple-comparison test (panels G-I). ****P $\leq$.0001; ***P $\leq$.001; **P $\leq$.01; *P $\leq$.05. CHX, cycloheximide; TNFR1/2, TNF- α receptor 1/2.

elevated blood counts and molecular remissions characterized by reduced Jak2VF allelic burden in patients with PV receiving IFN- α therapy.³⁵⁻³⁸ Despite these promising clinical outcomes, the precise mechanisms underlying IFN- α efficacy have remained elusive. Here, we show that NK cells are indispensable for the efficacy of ropegIFN- α -2b therapy in PV, providing

novel mechanistic insight into how type-I interferon-mediated immune modulation contributes to therapeutic efficacy in MPN.

Type-I interferons are pleiotropic cytokines inducing multifaceted antitumor immune responses³⁹ and can directly act on

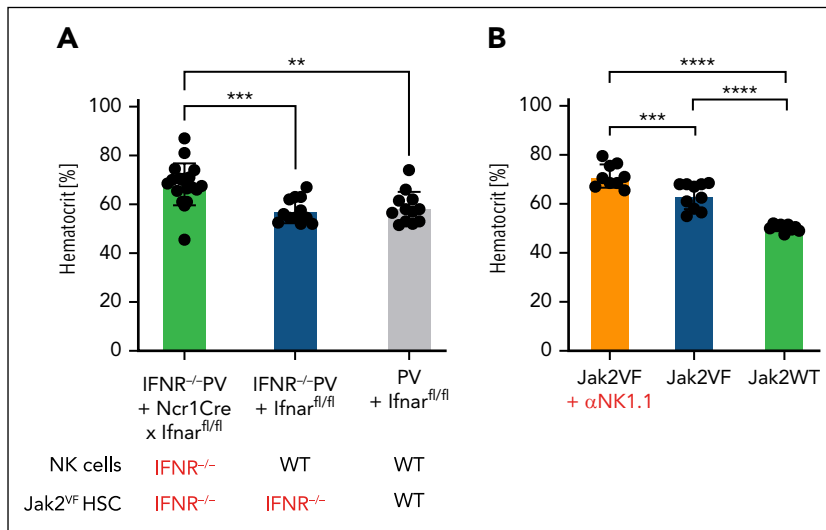


Figure 6. PV BMC in which solely NK cells are deficient for type-I interferon receptor expression show accelerated disease progression. (A) BMC were generated by transplanting lethally irradiated C57BL/6 mice with a mixture of 15% Ifnar1^{-/-} × VavCre × Jak2^{VF} (IFN^{-/-} PV) and 85% Ncr1Cre × Ifnar^{fl/fl} BM (n = 19) or, as control, 15% IFN^{-/-} PV and 85% WT (Ifnar^{fl/fl}) BM (n = 12), or 15% VavCre × Jak2^{VF} (PV) and 85% WT (Ifnar^{fl/fl}) BM (n = 12). Hematocrit was determined 4 weeks after BM transplantation. (B) BMC were generated by transplanting lethally irradiated C57BL/6 mice with a mixture of 15% VavCre × Jak2V617F (Jak2VF) and 85% WT (CD45.1) BM (Jak2VF) or, as control, 15% WT and 85% WT (CD45.1) BM (Jak2WT). In the indicated groups, NK cells were concomitantly depleted by weekly injection with an antibody against NK1.1. Hematocrit was determined 4 weeks after BM transplantation. Horizontal bars represent mean ± SD. Statistical significance was determined with 1-way ANOVA followed by the Tukey multiple-comparison test. ****P ≤ .0001; ***P ≤ .001; **P ≤ .01.

tumor cells by inducing cell cycle arrest and apoptosis.³⁹ Proposed mechanisms of IFN-α action in MPN have primarily entailed direct effects on Jak2VF malignant HSC clones.

IFN-α induces transient proliferation of dormant HSCs, rendering them more susceptible to apoptosis and reducing the HSC pool through impaired self-renewal.^{40–42} Because IFN-α–induced cell cycle entry is more pronounced in Jak2VF malignant stem cells, preferential reduction of the malignant HSC pool has been proposed to explain decreased Jak2VF allelic burden in patients with PV treated with IFN-α.^{12,26} Although our data do not question these direct effects, they clearly demonstrate that NK cells are essential for the therapeutic effects of IFN-α, likely acting in synergy with these mechanisms.

NK cells are key innate immune effectors and play a central role in cancer immunosurveillance.⁴³ In myeloid malignancies, NK cells are often functionally impaired, which is associated with poor prognosis.^{44,45} Consistent with this, we show that PV mice lacking NK cells or IFNAR1 expression on NK cells exhibit accelerated disease progression, indicating that NK cells must directly sense type-I interferon to effectively restrain disease progression. Notably, even basal levels of type-I interferon, which are constitutively produced in the absence of infection or inflammation, were sufficient to attenuate disease progression, highlighting the fundamental role of type-I interferon in immune activation under steady-state conditions.³⁴ Thus, our data show an essential role of type-I interferon sensing on NK cells mediated immune surveillance of the hematopoietic compartment.

Several studies reported increased circulating NK cells in patients with Jak2VF positivity after IFN-α therapy, with a shift toward CD56^{bright} NK cells, a subset characterized by enhanced immunostimulatory and cytokine production capacity.^{30,31} Consistent with this, we observed increased CD56^{bright} NK cells in patients with PV or ET receiving IFN-α therapy. Interestingly, in humans, NK cell expansion was observed in the peripheral blood, in contrast to our observations in mice. It will therefore be interesting to explore whether CD56^{bright} NK

cell frequencies could serve as a surrogate marker for therapeutic efficacy.

Consistent with previous reports,⁴⁶ IFN-α therapy in mice stimulated NK cell proliferation independent of PV, increasing both frequency and absolute NK cell number in the BM. Notably, IFN-α specifically expanded CD27⁺ immature NK cells, which correspond to CD56^{bright} NK cells in humans. However, we did not see increased frequencies of NK cells in mice carrying the mutant allele, suggesting that malignant cells do not induce NK cell expansion. NK cell expansion was accompanied by phenotypic changes, including upregulation of the activating receptor NKG2D, whereas the inhibitory receptor NKG2A remained unchanged.

These findings suggest that both expansion and activation of NK cells contribute to therapeutic effects of IFN-α in PV. Although direct interferon signaling in NK cells is required, proliferation may also involve stimulation of other immune cells such as DCs, which then activate NK cells.⁴⁷ Although increased numbers of CD4⁺ T cells and NK cells have been reported in patients with chronic myeloid leukemia treated with IFN-α,^{48,49} raising the possibility that T cells contribute to therapeutic effects of IFN-α in PV. However, in our PV mouse model, depletion of either CD4⁺ or CD8⁺ T cells did not affect therapeutic efficacy of IFN-α, and no changes in T-cell frequencies were observed in our patient cohort.

Our findings further demonstrate that NK cells directly kill HSPCs, preferentially targeting Jak2VF-mutated stem cells. Although NKG2D and its ligand Rae-1 were upregulated upon IFN-α treatment, blocking NKG2D did not impair NK cell-mediated killing. In contrast, blocking TNF-α completely abolished selective killing, indicating that NK cell-mediated elimination of malignant HSCs depends on TNF-α signaling. TNF-α has been shown to synergize with IFN-γ to induce inflammatory cell death via the JAK/STAT1/interferon regulatory factor 1 axis.⁵⁰ Given that Jak2 mediates IFN-γ receptor signaling,⁵¹ constitutive Jak2 activation in Jak2VF-mutant cells may increase susceptibility to TNF-α-mediated killing, promoting preferential elimination of malignant cells.

Although recent clinical phase 2 trials in MF and ET using programmed cell death protein 1 inhibitors such as nivolumab or pembrolizumab^{52,53} were terminated early because of limited efficacy and safety concerns, immune based therapeutic approaches in MPN remain of considerable interest. Current efforts include next-generation checkpoint inhibitors⁵⁴ and combination strategies with immune-permissive JAK inhibitors such as fedratinib.⁵⁵ Our findings suggest that NK activation represents a promising therapeutic strategy worthwhile of further exploration.⁵⁶

In summary, we present novel mechanistic insight into how immune stimulation by IFN- α contributes to its therapeutic effects in PV. NK cells are essential for disease progression, with IFN- α -mediated NK cell activation and NK cell-derived TNF- α identified as key mechanisms of disease control in PV. These findings highlight the potential of NK cell-based strategies to enhance immune surveillance and improve outcomes in MPN.

Acknowledgments

The authors thank A. Klaric for assistance and technical support, and the team of the central animal facility of the University of Mainz (Translational Animal Research Center) for expert animal care. They are grateful to PharmaEssentia for providing murine ropeginterferon- α (mP1101); the Tissue Biobank of the University Medical Center Mainz, for providing access to their technology platform; and Bonny Adami for her technical support. They also thank the fluorescence-activated cell sorting (FACS) Core Facility Flow Cytometry at the Paul Klein Centre for Immune Intervention and the Research Center for Immunotherapy of the University Medical Center of the Johannes Gutenberg University Mainz for providing support and instrumentation (FACSARIA III and FACSsymphony A5) funded by the Deutsche Forschungsgemeinschaft (DFG [German Research Foundation]) INST 371/44-1 FUGB and INST 371/45-1 FUGB.

This work was funded by the DFG (project 318346496-SFB1292, subproject TP21; M.P.R., S.M., and D.S.); subproject TP13 (H.C.P. and H.S.); subproject TP19N (F.M.); subproject TP22 (M.M.G.); subproject TPQ01 (M.M.G., A.H., and M.N.H.-W.); and subproject TP01 (T.B.); as well as KO2155/7-1 (project ID 428858786; S.K.). F.H.H. was supported by grants from the German Research Council (DFG; HE6233/15-1 and 16-1, project number 517204983 and HE6233/4-2, project number 320028127). The production of mP1101 is supported by PharmaEssentia Corporation.

Authorship

Contribution: J.L. performed research and statistical analysis, analyzed and interpreted data, and wrote the manuscript; T.M.S. and S.-J.L. contributed vital new reagents; F.M. analyzed and interpreted data, contributed analytical tools, and wrote the manuscript; N.A., Z.R., N.H., K.H., M.M.G., A.H., M.N.H.-W., C.T.M., K.S., and M.K. performed research and analyzed and interpreted data; T.B. and H.S. analyzed and interpreted data; C.C., F.H.H., and S.K. contributed vital new reagents and analyzed and interpreted data; D.S. contributed vital new reagents, performed research, and analyzed and interpreted data; and H.C.P., M.P.R., and S.M. designed and performed the research, analyzed and interpreted data, performed statistical analysis, and wrote the manuscript.

Conflict-of-interest disclosure: S.-J.L. works for PharmaEssentia Corporation, which develops the mP1101 described in this article. F.H.H. served as an advisor for Novartis, CTI BioPharma (CTI), Celgene/Bristol Myers Squibb (BMS), Janssen, AbbVie, GSK, Merck, and AOP Pharma; and received research funding from Novartis, Celgene/BMS and CTI. S.K. reports research grant/funding from Novartis, BMS, AOP Pharma, Imago Biosciences, Janssen, and Geron; reports advisory board honoraria and consulting fees from Pfizer, Incyte, Ariad, Novartis, AOP Pharma, BMS, Celgene, Geron, Janssen, CTI, Roche, Baxalta, Sanofi, Sierra Oncology, GSK, AbbVie, and Merck Sharp and Dohme (MSD); reports a patent for a bromodomain and extraterminal protein inhibitor at RWTH Aachen University; reports honoraria from Novartis, BMS, Celgene, Geron, Janssen, Pfizer, Incyte, Ariad, Shire, Roche, AOP Pharma, GSK, AbbVie, PharmaEssentia, iOMEDICO, and MPN Hub; and reports other financial support (eg, travel support) from Alexion, Novartis, BMS, Incyte, Ariad, AOP Pharma, Baxalta, CTI, Pfizer, Sanofi, Celgene, Shire, Janssen, Geron, AbbVie, Kartos, Sierra Oncology, Imago Biosciences, GSK, PharmaEssentia, MSD, and iOMEDICO. M.P.R. has consulted for, and received honoraria from, AOP Pharma, BMS, GSK, Sobi, Johnson & Johnson, Cogent, BeiGene, Lilly, and Merck. The remaining authors declare no competing financial interests.

ORCID profiles: T.M.S., 0000-0002-6351-4119; F.M., 0000-0003-3252-7758; N.A., 0000-0001-8357-0938; M.M.G., 0000-0003-1499-7772; A.H., 0009-0001-4547-3843; M.N.H.-W., 0000-0002-1131-8714; T.B., 0000-0002-3232-8065; H.S., 0000-0001-9462-3644; D.S., 0000-0003-0330-7959; F.H.H., 0000-0003-2438-1955; S.K., 0000-0002-1011-8171; H.C.P., 0000-0002-7926-0651; M.P.R., 0000-0002-3991-5721; S.M., 0009-0004-3026-6245.

Correspondence: Sabine Muth, Institute of Immunology, University Medical Center of the Johannes Gutenberg University Mainz, Langenbackstr 1, 55131 Mainz, Germany; email: sabine.muth@uni-mainz.de; and Markus P. Radsak, Department of Medicine VII, Hematology and Oncology, Donau-Isar-Klinikum Deggendorf, Perlaserger Str. 41, 94469 Deggendorf, Germany; email: markus.radsak@donau-isar-klinikum.de.

Footnotes

Submitted 9 March 2026; accepted 1 May 2026; prepublished online on *Blood* First Edition 2 June 2026. <https://doi.org/10.1182/blood.2026033951>.

*H.C.P., M.P.R., and S.M. are joint senior authors.

RNA sequencing data presented in this manuscript have been deposited in the Gene Expression Omnibus database (accession number GSE289788).

All data and protocols are available from the corresponding authors, Sabine Muth (sabine.muth@uni-mainz.de) and Markus P. Radsak (radsak@uni-mainz.de), on request.

The online version of this article contains a data supplement.

There is a *Blood* Commentary on this article in this issue.

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REFERENCES

- Perner F, Pahl HL, Zeiser R, Heidel FH. Malignant JAK-signaling: at the interface of inflammation and malignant transformation. *Leukemia*. 2025;39(5):1011-1030.
- Baxter EJ, Scott LM, Campbell PJ, et al. Acquired mutation of the tyrosine kinase JAK2 in human myeloproliferative disorders. *Lancet*. 2005;365(9464):1054-1061.
- Levine RL, Wadleigh M, Cools J, et al. Activating mutation in the tyrosine kinase JAK2 in polycythemia vera, essential thrombocythemia, and myeloid metaplasia with myelofibrosis. *Cancer cell*. 2005;7(4):387-397.
- James C, Ugo V, Le Couédic JP, et al. A unique clonal JAK2 mutation leading to constitutive signalling causes polycythaemia vera. *Nature*. 2005; 434(7037):1144-1148.
- Kralovics R, Passamonti F, Buser AS, et al. A gain-of-function mutation of JAK2 in myeloproliferative disorders. *N Engl J Med*. 2005;352(17):1779-1790.
- Ling VY, Heidel FH, Bywater MJ. Pathogenesis and management of high

- molecular risk myeloproliferative neoplasms. *Haematologica*. 2025;110(4):863-876.
7. Hasselbalch HC. Chronic inflammation as a promotor of mutagenesis in essential thrombocythemia, polycythemia vera and myelofibrosis. a human inflammation model for cancer development? *Leuk Res*. 2013; 37(2):214-220.
 8. Jayavelu AK, Schnöder TM, Perner F, et al. Splicing factor YBX1 mediates persistence of JAK2-mutated neoplasms. *Nature*. 2020; 588(7836):157-163.
 9. Kiladjian JJ, Cassinat B, Chevret S, et al. Pegylated interferon-alfa-2a induces complete hematologic and molecular responses with low toxicity in polycythemia vera. *Blood*. 2008;112(8):3065-3072.
 10. Quintás-Cardama A, Kantarjian H, Manshouri T, et al. Pegylated interferon alfa-2a yields high rates of hematologic and molecular response in patients with advanced essential thrombocythemia and polycythemia vera. *J Clin Oncol*. 2009;27(32): 5418-5424.
 11. Abu-Zeinah G, Silver RT, Abu-Zeinah K, Scandura JM. Normal life expectancy for polycythemia vera (PV) patients is possible. *Leukemia*. 2022;36(2):569-572.
 12. Mullally A, Bruedigam C, Poveromo L, et al. Depletion of Jak2V617F myeloproliferative neoplasm-propagating stem cells by interferon- α in a murine model of polycythemia vera. *Blood*. 2013;121(18): 3692-3702.
 13. Hasan S, Lacout C, Marty C, et al. JAK2V617F expression in mice amplifies early hematopoietic cells and gives them a competitive advantage that is hampered by IFN α . *Blood*. 2013;122(8):1464-1477.
 14. Perner F, Perner C, Ernst T, Heidel FH. Roles of JAK2 in aging, inflammation, hematopoiesis and malignant transformation. *Cells*. 2019;8(8):854.
 15. Plataniias LC. Mechanisms of type-I- and type-II-interferon-mediated signalling. *Nat Rev Immunol*. 2005;5(5):375-386.
 16. van Boxel-Dezaire AHH, Rani MRS, Stark GR. Complex modulation of cell type-specific signaling in response to type I interferons. *Immunity*. 2006;25(3):361-372.
 17. Mullally A, Lane SW, Ball B, et al. Physiological Jak2V617F expression causes a lethal myeloproliferative neoplasm with differential effects on hematopoietic stem and progenitor cells. *Cancer cell*. 2010;17(6): 584-596.
 18. Georgiades P, Ogilvy S, Duval H, et al. VavCre transgenic mice: a tool for mutagenesis in hematopoietic and endothelial lineages. *Genesis (New York, NY : 2000)*. 2002;34(4):251-256.
 19. Shen FW, Saga Y, Litman G, et al. Cloning of Ly-5 cDNA. *Proc Natl Acad Sci U S A*. 1985; 82(21):7360-7363.
 20. Narni-Mancinelli E, Chaix J, Fenis A, et al. Fate mapping analysis of lymphoid cells expressing the NKp46 cell surface receptor. *Proc Natl Acad Sci U S A*. 2011;108(45): 18324-18329.
 21. Kamphuis E, Junt T, Waibler Z, Forster R, Kalinke U. Type I interferons directly regulate lymphocyte recirculation and cause transient blood lymphopenia. *Blood*. 2006;108(10): 3253-3261.
 22. Müller U, Steinhoff U, Reis LF, et al. Functional role of type I and type II interferons in antiviral defense. *Science*. 1994;264(5167):1918-1921.
 23. Muth S, Klaric A, Radsak M, Schild H, Probst HC. CD27 expression on Treg cells limits immune responses against tumors. *J Mol Med*. 2022;100(3):439-449.
 24. Sakatoku K, Nakashima Y, Nagasaki J, et al. Immunomodulatory and direct activities of ropeginterferon alfa-2b on cancer cells in mouse models of leukemia. *Cancer Sci*. 2022;113(7):2246-2257.
 25. Heidel FH, Bullinger L, Feng Z, et al. Genetic and pharmacologic inhibition of β -catenin targets imatinib-resistant leukemia stem cells in CML. *Cell Stem Cell*. 2012;10(4):412-424.
 26. Austin RJ, Straube J, Bruedigam C, et al. Distinct effects of ruxolitinib and interferon- α on murine JAK2V617F myeloproliferative neoplasm hematopoietic stem cell populations. *Leukemia*. 2020;34(4): 1075-1089.
 27. Becht E, McInnes L, Healy J, et al. Dimensionality reduction for visualizing single-cell data using UMAP. *Nat Biotechnol*. 2018;37(1):38-44.
 28. Chiossone L, Chaix J, Fuseri N, Roth C, Vivier E, Walzer T. Maturation of mouse NK cells is a 4-stage developmental program. *Blood*. 2009;113(22):5488-5496.
 29. Ng ZY, Fuller KA, Mazza-Parton A, Erber WN. Morphology of myeloproliferative neoplasms. *Int J Lab Hematol*. 2023; 45(suppl 2):59-70.
 30. Riley CH, Brimnes MK, Hansen M, et al. Interferon- α induces marked alterations in circulating regulatory T cells, NK cell subsets, and dendritic cells in patients with JAK2V617F-positive essential thrombocythemia and polycythemia vera. *Eur J Haematol*. 2016;97(1):83-92.
 31. Riley CH, Hansen M, Brimnes MK, et al. Expansion of circulating CD56bright natural killer cells in patients with JAK2-positive chronic myeloproliferative neoplasms during treatment with interferon- α . *Eur J Haematol*. 2015;94(3):227-234.
 32. Abel AM, Yang C, Thakar MS, Malarkannan S. Natural killer cells: development, maturation, and clinical utilization. *Front Immunol*. 2018;9:1869.
 33. McNab F, Mayer-Barber K, Sher A, Wack A, O'Garra A. Type I interferons in infectious disease. *Nat Rev Immunol*. 2015;15(2): 87-103.
 34. Schaupp L, Muth S, Rogell L, et al. Microbiota-induced type I interferons instruct a poised basal state of dendritic cells. *Cell*. 2020;181(5):1080-1096.e19.
 35. Vachhani P, Mascarenhas J, Bose P, et al. Interferons in the treatment of myeloproliferative neoplasms. *Ther Adv Hematol*. 2024;15:20406207241229588.
 36. Hasselbalch HC, Holmström MO. Perspectives on interferon- α in the treatment of polycythemia vera and related myeloproliferative neoplasms: minimal residual disease and cure? *Semin Immunopathol*. 2019;41(1):5-19.
 37. Healy FM, Dahal LN, Jones JRE, Floisand Y, Woolley JF. Recent progress in interferon therapy for myeloid malignancies. *Front Oncol*. 2021;11:769628.
 38. Kiladjian JJ, Giraudeau S, Cassinat B. Interferon- α for the therapy of myeloproliferative neoplasms: targeting the malignant clone. *Leukemia*. 2016;30(4): 776-781.
 39. Parker BS, Rautela J, Hertzog PJ. Antitumour actions of interferons: implications for cancer therapy. *Nat Rev Cancer*. 2016;16(3): 131-144.
 40. Essers MAG, Offner S, Blanco-Bose WE, et al. IFN α activates dormant haematopoietic stem cells in vivo. *Nature*. 2009;458(7240):904-908.
 41. Sato T, Onai N, Yoshihara H, Arai F, Suda T, Ohteki T. Interferon regulatory factor-2 protects quiescent hematopoietic stem cells from type I interferon-dependent exhaustion. *Nat Med*. 2009;15(6):696-700.
 42. Rao TN, Hansen N, Stetka J, et al. JAK2-V617F and interferon- α induce megakaryocyte-biased stem cells characterized by decreased long-term functionality. *Blood*. 2021;137(16): 2139-2151.
 43. Malmberg KJ, Carlsten M, Björklund A, Sohlberg E, Bryceson YT, Ljunggren HG. Natural killer cell-mediated immunosurveillance of human cancer. *Semin Immunol*. 2017;31:20-29.
 44. Carlsten M, Järås M. Natural killer cells in myeloid malignancies: immune surveillance, NK cell dysfunction, and pharmacological opportunities to bolster the endogenous NK cells. *Front Immunol*. 2019;10:2357.
 45. Naismith E, Steichen J, Sopper S, Wolf D. NK cells in myeloproliferative neoplasms (MPN). *Cancers*. 2021;13(17):4400.
 46. Biron CA, Nguyen KB, Pien GC, Cousens LP, Salazar-Mather TP. Natural killer cells in antiviral defense: function and regulation by innate cytokines. *Annu Rev Immunol*. 1999; 17:189-220.
 47. Lucas M, Schachterle W, Oberle K, Aichele P, Diefenbach A. Dendritic cells prime natural killer cells by trans-presenting interleukin 15. *Immunity*. 2007;26(4):503-517.
 48. Ilander M, Kreutzman A, Rohon P, et al. Enlarged memory T-cell pool and enhanced Th1-type responses in chronic myeloid leukemia patients who have successfully

- discontinued IFN- α monotherapy. *PLoS One*. 2014;9(1):e87794.
49. Ureshino H, Kamachi K, Kidoguchi K, Kimura S. IFN- α treatment may enable discontinuation of TKIs in NK cell-licensed patients with CML-CP. *EJHaem*. 2024;5(6): 1278-1282.
 50. Karki R, Sharma BR, Tuladhar S, et al. Synergism of TNF- α and IFN- γ triggers inflammatory cell death, tissue damage, and mortality in SARS-CoV-2 infection and cytokine shock syndromes. *Cell*. 2021;184(1): 149-168.e17.
 51. Schneider WM, Chevillotte MD, Rice CM. Interferon-stimulated genes: a complex web of host defenses. *Annu Rev Immunol*. 2014; 32:513-545.
 52. Abou Dalle I, Kantarjian H, Daver N, et al. Phase II study of single-agent nivolumab in patients with myelofibrosis. *Ann Hematol*. 2021;100(12):2957-2960.
 53. Hobbs G, Cimen Bozkus C, Moshier E, et al. PD-1 inhibition in advanced myeloproliferative neoplasms. *Blood Adv*. 2021;5(23):5086-5097.
 54. Yadav R, Hakobyan N, Wang JC. Role of next generation immune checkpoint inhibitor (ICI) therapy in Philadelphia negative classic myeloproliferative neoplasm (MPN): review of the literature. *Int J Mol Sci*. 2023;24(15): 12502.
 55. Isfort S, von Bubnoff N, Al-Ali HK, et al. FRACTION: protocol of a phase II study of Fedratinib and Nivolumab combination in patients with myelofibrosis and resistance or suboptimal response to JAK-inhibitor treatment of the German MPN Study Group (GSG-MPN). *Ann Hematol*. 2024;103(8): 2775-2785.
 56. Miyazato K, Hayakawa Y. Pharmacological targeting of natural killer cells for cancer immunotherapy. *Cancer Sci*. 2020;111(6): 1869-1875.

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